

Geometry of Splats

Classical WFR theory: minimize $\mathcal{F} : \mathcal{P}_{2,ac}(\mathbb{R}^d) \rightarrow \mathbb{R}$ such as,

- Potential energy: $\mathcal{F}(\mu) = \int V(x) \mu(dx)$
- Internal energy: $\mathcal{F}(\mu) = \int U(\mu(x)) dx$
- Interaction energy: $\mathcal{F}(\mu) = \int K(x, y) \mu(dx) \mu(dy)$
- Ex. divergences: $\mathcal{F}(\mu) = \text{KL}(\mu \mid \pi), \chi^2(\mu \mid \pi), \dots$

Splats: minimize $\mathcal{R}(\mu) := \mathbb{E}_{x,y} \left[\|\mathcal{A}[f_\mu](x) - y\|^2 \right]$

Geometry of Splats

Splats functions are *mean-field two layer neural networks!*

Some unique phenomenology:

- Highly sparse, localized activation patterns.
- Neurons are smoothed particles. Can teleport, prune, etc.